



Student Preparation Program for the Semester 2

Programming Tutorial

IT25100310

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First Year Division

Question 1

- a) Declare two integer variables, miles and yards, and one double variable for kilometers
- b) Initialize the variables to hold the number of miles and yards in a marathon, respectively (miles to 26 and yards to 385).
- c) Write an expression to calculate kilometers from miles and yards.
- d) Save the result of the expression in the variable kilometers. a. Note: One mile is 1.609 kilometers. There are 1760.0 yards in a mile

Code

```
1 //Question 1
2
3 public class it25100310Q1{
4     public static void main (String[]args){
5
6         //Declare variables
7         int miles = 26;
8         int yards = 385;
9         double kilometers = 0;
10
11         kilometers = (miles*1.609) + (yards/1760.0)*1.609;
12         System.out.print("\n26m 385y equal to : "+kilometers+" KM\n");
13     }
14 }
15
16
```

Out put

```
C:\Users\lasit\Desktop\2026 recapIP>java it25100310Q1.java
26m 385y equal to : 42.18596875 KM
```

Question 2

Write a Java program to add the elements of arrays A and B and store the result in a new array,

int A[5] = {10, 20, 30, 40, 50};

int B[5] = {34, 67, 12, 89, 12};

int C[5];

Calculate A+B and store in Array C. → Compute the sum of corresponding elements in arrays A and B and store them in array C.

Example Output: Array C: [44, 87, 42, 129, 62]

Code

```
it25100310Q2.java
1 public class it25100310Q2 {
2     public static void main (String[] args) {
3
4         //Define
5         int[] A = {10, 20, 30, 40, 50};
6         int[] B = {34, 67, 12, 89, 12};
7         int[] C = new int[5];
8
9         //add sum to each index
10        for (int i = 0; i < 5; i++) {
11            C[i] = A[i] + B[i];
12        }
13
14        //print C Array
15        System.out.print("Array - ");
16        for (int i = 0; i < 5; i++) {
17            System.out.print(C[i]);
18            if (i < (5-1)) {
19                System.out.print(", ");
20            }
21        }
22        System.out.print("\n");
23    }
24 }
25 }
```

Out put

```
C:\Users\lasit\Desktop\2026 recapIP>java it25100310Q2.java
Array - [44,87,42,129,62]
```

Question 3

a) Create a class called **EvenOddNumber**.

b) It has a method called **findEvenOrOdd(int i)** to find whether a number is even or odd.

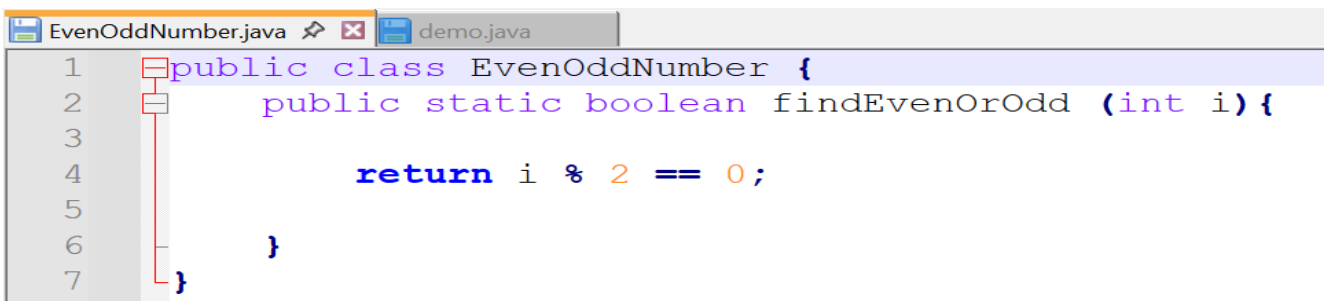
c) This method takes an integer argument and returns a **boolean** value.

(Note: There is a data type called boolean in java which can store true/false values)

d) Then create another class Demo with the main method and create an object from **EvenOddNumber** class.

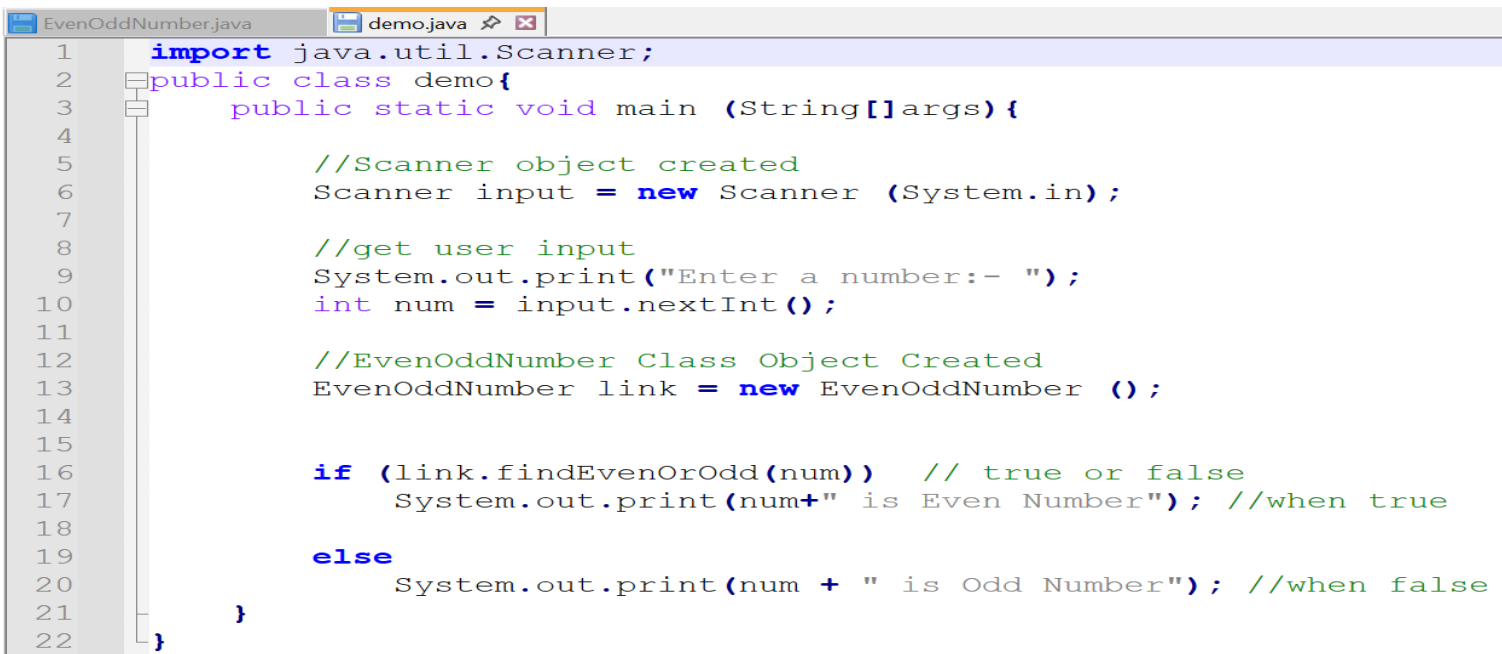
e) Using the created object, call the **findEvenOrOdd()** method and display outputs indicating whether those are even numbers or odd numbers.

user-defined method



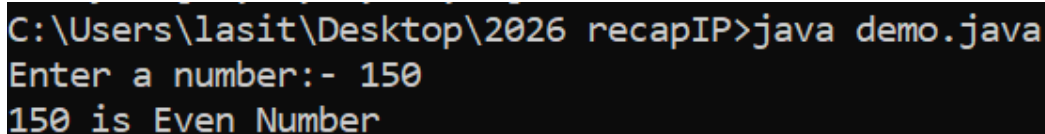
```
1 public class EvenOddNumber {
2     public static boolean findEvenOrOdd (int i){
3
4         return i % 2 == 0;
5
6     }
7 }
```

main method



```
1 import java.util.Scanner;
2 public class demo{
3     public static void main (String[] args){
4
5         //Scanner object created
6         Scanner input = new Scanner (System.in);
7
8         //get user input
9         System.out.print("Enter a number:- ");
10        int num = input.nextInt();
11
12        //EvenOddNumber Class Object Created
13        EvenOddNumber link = new EvenOddNumber ();
14
15
16        if (link.findEvenOrOdd(num)) // true or false
17            System.out.print(num+" is Even Number"); //when true
18
19        else
20            System.out.print(num + " is Odd Number"); //when false
21    }
22 }
```

Out put



```
C:\Users\lasit\Desktop\2026 recapIP>java demo.java
Enter a number:- 150
150 is Even Number
```

Question 4

Write a program that draws the following figures one above the other.

Using while loop

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

Using for loop

```
      *
     * *
    * * *
   * * * *
  * * * * *
```

```
drawing.java
1 public class drawing {
2     public static void main(String[] args) {
3
4         int i, j, s;
5
6         // Using while loop
7         i = 0;
8         while (i < 5) {
9             j = 0;
10            while (j < 5) {
11                System.out.print("* ");
12                j++;
13            }
14            System.out.println();
15            i++;
16        }
17
18        System.out.println(); // empty line between patterns
19
20
21
22        //Using for loop
23        int row = 5;
24        for (i=0;i<=5;i++){
25
26            for (s=0;s<=row-i;s++){
27                System.out.print(" ");
28            }
29
30            for (j=0;j<i;j++){
31                System.out.print("* ");
32            }
33            System.out.println(); // next line
34        }
35
36    }
37 }
```

Out put

```
C:\Users\lasit\Desktop\2026 recapIP>java drawing.java
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

      *
     * *
    * * *
   * * * *
  * * * * *
```

Question 5

Please submit your answer to the question below to Git. When committing the file, ensure that you include a meaningful commit message (e.g., "Lab01- Question05").

Create a class called "Calculator" and implement three Java methods to perform the following operations.

- add()– add two integers pass as parameters and return the result
- multiply()– multiply two integers pass as parameters and return the result
- square()– receive an integer as a parameter and return the result after multiplying the number by itself. Use the above methods in the Main Method to calculate the result of the following mathematical expressions:

1. $(3 * 4 + 5 * 7)^2$

2. $(4 + 7)^2 + (8 + 3)^2$

Code

```
Calculator.java
1 public class Calculator{
2
3 public static int add (int a,int b){
4     return a+b;
5 }
6
7 public static int multiply (int a,int b){
8     return a*b;
9 }
10
11 public static int square (int a){
12     return a*a;
13 }
14 }
15
16 public static void main (String[] args){
17
18     System.out.println( "(3*4+5*7)\u00B2 = "+square( add(multiply(3,4),multiply(5,7)))));
19     System.out.print( "(4+7)\u00B2 + (8+3)\u00B2 = "+add(square( add (4,7)), square(add(8,3))));
20
21 }
22 }
```

Out put

```
C:\Users\lasit\Desktop\2026 recapIP>java Calculator.java
(3*4+5*7)2 = 2209
(4+7)2 + (8+3)2 = 242
```

Git Link

[IT25100310/it1120-Semester-2-Preparation-IT25100310Q5: Java solution for Lab 01 Question 05. Implements a Calculator class with add, multiply, and square methods. The program evaluates two expressions: \$\(3 \times 4 + 5 \times 7\)^2\$ and \$\(4 + 7\)^2 + \(8 + 3\)^2\$. Demonstrates basic method usage, arithmetic operations, and clean modular code structure.](#)